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Performance analysis of Carnot battery pumped thermal electricity storage aided with concentrated solar thermal input

Mengqi Du [a], He Yang [a], Xiaoze Du [a,b,*], Hongwei Wu [c]

[a] Key Laboratory of Power Station Energy Transfer Conversion and System, North China Electric Power University, Ministry of Education, Beijing, 102206, China
[b] School of Energy and Power Engineering, Lanzhou University of Technology, Lanzhou, 730050, China
[c] School of Physics, Engineering and Computer Science, University of Hertfordshire, Hatfield, AL109AB, UK

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ABSTRACT

The integration of pumped thermal electricity storage (PTES) with concentrated solar power (CSP) systems offers significant potential to enhance renewable energy consumption, optimize solar energy utilization, and elevate the overall efficiency and performance of PTES systems. Existing research on system integration primarily focuses on CSP plants utilizing molten salt for heat collection and PTES systems employing liquid-based heat storage. Limited studies have been conducted on integrating PTES systems incorporating solid packed beds with CSP plants utilizing alternative heat collection materials. For PTES systems featuring solid packed beds and CSP plants employing solid particle heat collection, this study proposes an innovative CSP-aided PTES system (CSP-PTES) that harnesses solar thermal input from a concentrated heliostat field. A thermodynamic model was developed, and mathematical models of system performance criteria were obtained. The effects of critical design and component parameters on system performance were investigated. Employing the Non-dominated Sorting Genetic Algorithm II (NSGA-II), optimization and the trade-off analysis of round-trip efficiency, energy density and power density were performed. The results demonstrate that there is a corresponding decrease in both energy density and power density with round-trip efficiency. The introduction of solar thermal greatly improves the output energy and power density of the system, with solar efficiency reaching up to 48 %, while the round trip efficiency of the system is slightly reduced due to solar efficiency. The variation in energy and power densities is much larger than the round-trip efficiency, resulting in the optimal points with high energy and power densities, and low round-trip efficiency when the LINMAP decision is applied. However, the round-trip efficiency still managed to be about 58 %.

Nomenclature

BES	battery energy storage
CAES	compressed air energy storage
C	heat capacity, J·K^{-1} s^{-1}
c_p	specific heat capacity, J·K^{-1} kg^{-1}
CR	cold reservoir
CSP	concentrated solar power
CSP-PTES	CSP-PTES system
E_{solar}	solar energy, MW
E_X	solar exergy input, MW
e_X	specific exergy, MJ/kg
f_p	fractional pressure loss of heat exchanger
h	enthalpy, MJ/kg

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HR	hot reservoir
HX	heat exchanger
LINMAP	Linear Programming Techniques for Multidimensional Analysis of Preference
M	molar mass, kg·mol^{-1}
m	mass flow rate, kg/s
NSGA-II	Non-dominated Sorting Genetic Algorithm II
p	pressure, MPa
PHES	pumped hydroelectric energy storage
PTES	pumped thermal electricity storage
Q	heat transfer rate
R	molar gas constant, J·K^{-1} mol^{-1}
s-CO2	supercritical carbon dioxide
T	temperature, K

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* Corresponding author. Key Laboratory of Power Station Energy Transfer Conversion and System, North China Electric Power University, Ministry of Education, Beijing, 102206, China.
E-mail address: duxz@ncepu.edu.cn (X. Du).

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(continued)

t	time, s
TES	thermal energy storage
V	volume, m^3
W	power, MW
Greek Symbols	
β	compression/expansion ratio of the compressor/expander
γ	ratio of gas specific heat capacities
ε	efficiency of the heat exchanger
η	polytropic efficiency of the compressor/expander
η_{solar}	solar energy utilization efficiency
κ	parameter, $(\gamma-1)/\gamma$
ζ	fractional pressure loss of solar heat exchanger
ρ	density, $kg{\cdot}m^{-3}$
ρ_e	energy density, $kWh{\cdot}m^{-3}$
ρ_p	power density, $kW{\cdot}m^{-3}$
φ	porosity of the packed bed
χ	round-trip efficiency
Subscripts	
c	compressor
ch	charge
ch,net	net power input in charging process
dis	discharge
dis,net	net power output in discharging process
e	expander
end	at the end
env	environment
ex	external heat exchangers
g	gas
He	helium
HT	high temperature
in	at the inlet
ini	initiate
LT	low temperature
max	maximum
min	minimum
opt	optimal
out	at the outlet
re	regenerator
s	Solid or solid particle

1. Introduction

The escalating climate crisis and energy shortages have driven a substantial increase in the global installed capacity of renewable energy, now surpassing 3900 GW [1]. However, the renewable energy, including that of wind and solar power, are intermittent and variable, posing significant challenges to power grid stability [2]. The energy storage has emerged as a crucial technology for ensuring the safe integration of renewable energy into the grid [3–6]. However, among the large-scale energy storage technologies suitable for grid integration, the pumped hydroelectric energy storage (PHES) [7] and compressed air energy storage (CAES) [8] are heavily constrained by geographical limitations. The battery energy storage (BES) [9] faces significant challenges, including that of high costs, safety concerns, limited lifespan and post-processing.

The pumped thermal electricity storage (PTES) based on the reversible thermodynamic cycle, which can be classified into Carnot battery, has gained substantial attention owing to their geographical flexibility, cost-effectiveness, and scalability [10,11]. The fundamental principle of PTES entails transferring heat from the low-temperature side to the high-temperature side through a heat pump cycle, thus converting electrical energy into thermal energy. During the discharge phase, the stored heat is reconverted into electrical energy via a heat engine cycle [12].

Recently, extensive researches have been conducted on PTES. McTigue et al. [13] conducted a comprehensive analysis and optimization of PTES system parameters, determining that the efficiency of PTES systems is primarily contingent upon the performance of compressors and expanders. White et al. [14] delivered an in-depth analysis of thermodynamic losses in PTES reservoirs, indicating that these losses are correlated with operating temperature, reservoir geometry, and operational mode. McTigue et al. [15] and Laughlin [16] identified that PTES systems utilizing Brayton cycles could leverage gas-liquid heat exchangers and liquid heat storage media as alternatives to solid packed beds. Zhao et al. [17] conducted both conventional and advanced exergy analyses on PTES incorporating liquid storage, examining the components accountable for critical exergy losses within the system. Wang et al. [18] analyzed and optimized the discrete processes of charging, storing, and discharging in PTES systems, demonstrating that with reasonable temperature and pressure losses, a round-trip efficiency of 59 % and an energy density of 60 kWh/m^3 can be attained at a maximum temperature of 1100K. Ameen et al. [19] established the world's inaugural grid-scale PTES demonstration system and conducted extensive testing on the system's packed-bed thermal storage components. The findings revealed that stratified packed beds exhibit superior thermal storage performance and lower pressure loss compared to conventional packed beds. Ameen et al. [20] further examined the fundamental performance metrics of the PTES demonstration system, developed predictive mathematical models, and evaluated the system's operational performance, demonstrating that load-balancing cycles achieve higher efficiency than single storage cycles.

In terms of economic viability, Smallbone et al. [21] analyzed the levelized cost of PTES, concluding that PTES was cost-competitive with CAES and PHES, as well as offered complete locational flexibility. PTES boasts a low unit capacity cost, rendering it a cost-effective solution for both long-term and short-term storage needs. Martinek et al. [22] evaluated the optimal day-ahead scheduling and annual value of a dual-tank molten salt Joule-Brayton PTES system, assessing its economic performance across various scenarios. The aforementioned studies encompass both PTES systems utilizing solid packed beds and those employing liquid-based storage. However, they predominantly focus on evaluating the standalone performance of PTES systems, with limited attention given to their integration with other energy sources or systems.

The concentrated solar power (CSP) is an advanced solar energy technology that converts solar energy into high temperature heat and is subsequently utilized to produce electricity via a heat engine [23]. The next-generation CSP employs supercritical CO_2 Brayton cycles offer the advantages of a compact structure and high cycle efficiency. Nevertheless, the intermittent and variable nature of solar radiation poses significant challenges for CSP plants in fully utilizing solar energy [24]. Integrating a Brayton cycle PTES system with CSP plants can significantly enhance solar energy absorption and bolster system stability. Furthermore, the integration of PTES systems with CSP plants has the potential to share key system components, thereby reducing overall system investment costs.

Farres-Antunez et al. [25] investigated this potential and found that the round-trip efficiency of the two proposed integrated systems was below 60 %. McTigue et al. [26,27] proposed an integration scheme that combines sCO_2 recompression PTES cycles with CSP, demonstrating that CSP cycles integrated with PTES offer enhanced flexibility and can provide valuable grid energy storage services. Yang et al. [28] integrated solar input with PTES systems and investigated the impact of key design parameters on system performance and economic viability. Previous studies have predominantly focused on the integration of PTES systems utilizing liquid-based storage with solar energy. Notably, Petrollese et al. [29,30] proposed a pioneering scheme integrating a solid packed bed PTES system with CSP and investigated its performance under nominal conditions. Nevertheless, the majority of existing research has concentrated on PTES systems with liquid-based storage, leaving the integration of solid-state storage PTES systems with CSP comparatively underexplored. Moreover, molten salt has typically been employed as the heat collection material for CSP power plants in previous studies, while solid particles that can reach higher collector temperatures have not been considered, which is one of the crucial directions for the future development of CSP plants.

To further investigate the integration of PTES systems employing solid-state storage and CSP plants, several design options for the integration of PTES systems with CSP power plants are firstly proposed, and then one of the options is selected in this study to propose an innovative CSP-assisted PTES system (CSP-PTES) for CSP power plants with solid particle heat collection, which employs solid packed beds for energy storage and adds solar thermal input. In this case, solar heat is delivered via high-temperature particles sourced from a concentrated heliostat field, while the system's power input is supplied by the electrical grid or alternative renewable energy sources. The CSP-PTES system was designed with a nominal electric power input of 10 MW and a nominal storage capacity of 4 equivalent hours. The effects of major design parameters and component performance parameters on system performance are comprehensively explored. A multi-objective optimization of the system performance parameters based on the Non-dominated Sorting Genetic Algorithm II (NSGA-II) is conducted and the trade-offs between the main system performance metrics are determined. Furthermore, the optimal point of the Pareto fronts for the system performance metrics is determined based on the LINMAP decision-making method. This study investigates and optimizes the performance of the proposed CSP-PTES system, which is important for the design and application of PTES systems integrated with solar thermal.

2. System design and modeling

This study proposed an innovative CSP-aided PTES system (CSP-PTES) that incorporates solar heat input via a solid particles packed bed. This section offers a comprehensive overview of the CSP-PTES system. Based on this foundation, a thermodynamic model and performance evaluation metrics were established for subsequent parameter analysis.

The present CSP-PTES features a nominal input electric power of 10 MW and a storage capacity equivalent to 4 h of operation. The solar thermal energy is sourced from the CSP plant's solar collectors, whereas the electrical input is derived from the grid or other renewable energy sources.

2.1. Overview and layout of the CSP-PTES system

The integration of PTES with CSP plants primarily entails incorporating solar heat input from CSP systems. As shown in Fig. 1, there are three main methods for introducing solar heat input into PTES systems: 1) During the charging phase, the compressed high-temperature working fluid is further heated by solar energy before entering the hot reservoir, thereby significantly increasing its temperature and enhancing the energy storage density. 2) Solar heat energy is directly input into the hot reservoir to increase the stored energy, thereby providing a longer discharge duration. 3) During the discharge phase, the working fluid absorbs heat from the hot reservoir and subsequently absorbs solar heat input from a solar heat exchanger, thereby increasing its temperature. This method incorporates solar thermal energy solely during the discharging phase, thus having no impact on energy storage density.

The CSP-PTES system proposed in this study advances the traditional PTES system by incorporating a regenerator and solar thermal input. The schematic diagram of the system design is presented in Fig. 2. Here, HR and CR represent the hot and cold reservoirs, respectively, both comprising solid packed beds and tanks. The solar heat exchanger is a gas-solid fluidized bed heat exchanger with two tanks for storing solid particles before and after the heat exchange process. HX1, HX2, and HX3 denote external heat exchangers, which are employed to release excess heat generated during the charging and discharging cycles to the environment. Farres et al. [25] demonstrated that the inclusion of a recuperator with a temperature higher than ambient can achieve optimal round-trip efficiency for the PTES system. Additionally, the recuperator significantly diminishes the temperature gradient within the reservoir, resulting in a thinner thermal stratification layer, thereby

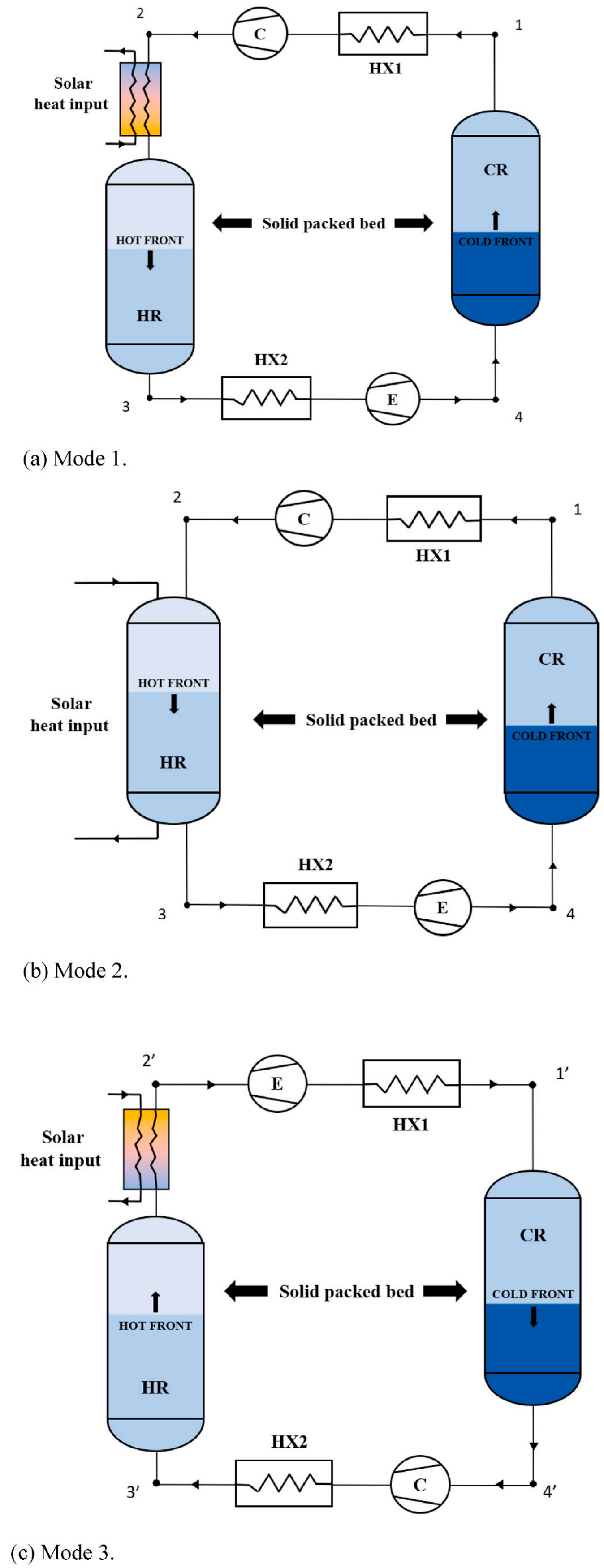


Fig. 1. The way PTES integrate solar thermal energy.

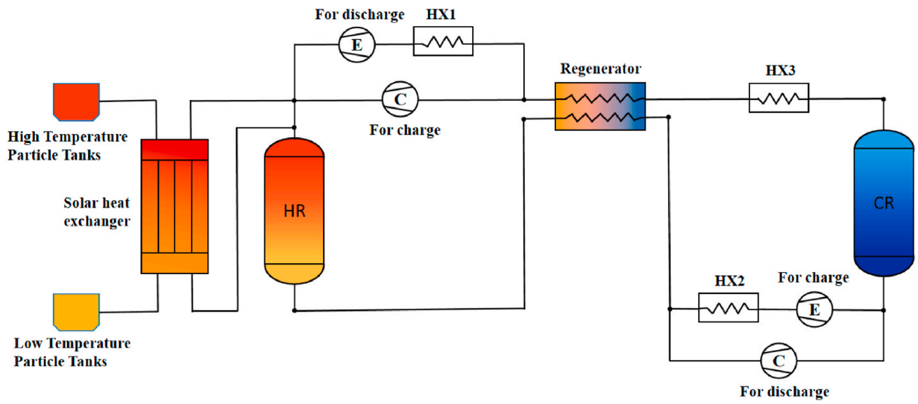


Fig. 2. CSP-PTES system design diagram.

enhancing the reservoir's thermal storage efficiency. The integration of solar thermal energy elevates the working temperature and electric power output of the heat engine cycle, thereby further enhancing the system's performances.

Fig. 3 shows the process flow diagram of the CSP-PTES system during charging and discharging, along with the corresponding T-s diagrams. The flow diagram for the charging cycle is illustrated in Fig. 3(a), where the arrows denote the flow direction of the working fluid during charging. The working fluid is first compressed by the compressor and directed into the hot reservoir for heat storage. It then passes through the regenerator and external heat exchanger to release heat before being expanded by the expander to reach a low-temperature state and entering the cold reservoir for cold storage. Finally, it traverses the regenerator once more, completing the cycle. In contrast, as illustrated in Fig. 3(b), the flow direction of the working fluid is reversed during the discharging process. Upon absorbing heat from the hot reservoir, the working fluid flows through the solar heat exchanger to gain additional thermal energy before undergoing expansion in the expander to produce electrical power.

This study employs a solid packed bed storage reservoir in the thermal energy storage section. The basalt particles demonstrate excellent thermal capacity and stability within the temperature range of −196 °C–800 °C. Consequently, the basalt particles are chosen as the TES material for the packed beds in both the hot and cold reservoirs in this study. Typical working fluids for the cycle include argon, helium, nitrogen, and supercritical carbon dioxide (sCO_2). Among these, monoatomic gases typically exhibit lower pressures at equivalent charging temperature ratios, thereby alleviating the pressure on the solid packed-bed reservoir. Helium, when compared to argon, demonstrates reduced heat transfer pressure losses and enhanced stability in output power [31]. Therefore, helium is selected as the working fluid in this study.

There are three primary methods of heat exchange between solid particles and gases: fixed bed, moving bed, and fluidized bed [32,33]. Due to its markedly superior heat transfer performance compared to fixed and moving beds, the fluidized bed is predominantly used for heat exchange between gases and solid particles [34]. This study utilizes a counterflow fluidized bed heat exchanger for heat transfer between high-temperature particles and the working fluid, selecting silicon carbide particles for their superior heat collection performance and availability.

2.2. Thermodynamic models

In the CSP-PTES system, the compressor and expander are modeled by incorporating polytropic efficiencies, η_c and η_e, to account for the irreversible losses associated with the compression and expansion processes.

For the compressor [35],

$$\frac{T_{\mathrm{c,out}}}{T_{\mathrm{c,in}}} = {\beta_{\mathrm{c}}}^{\kappa/\eta_c} \tag{1}$$

For the expander [35],

$$\frac{T_{\mathrm{e,out}}}{T_{\mathrm{e,in}}} = {\beta_{\mathrm{e}}}^{-\kappa\eta_e} \tag{2}$$

where β_{c} and β_{e} are the compression and expansion ratios, and κ is defined as κ=(γ-1)/γ, with γ being the specific heat ratio ($c_{\mathrm{p}}/c_{\mathrm{v}}$) [31].

In this paper, β_{c} and β_{e} refer to the compression and expansion ratios during the charging process, whereas parameters denoted with a prime '′' signify those relevant to the discharging process. Detailed parameters are provided in Table 1.

2.3. Heat transfer in packed bed reservoir

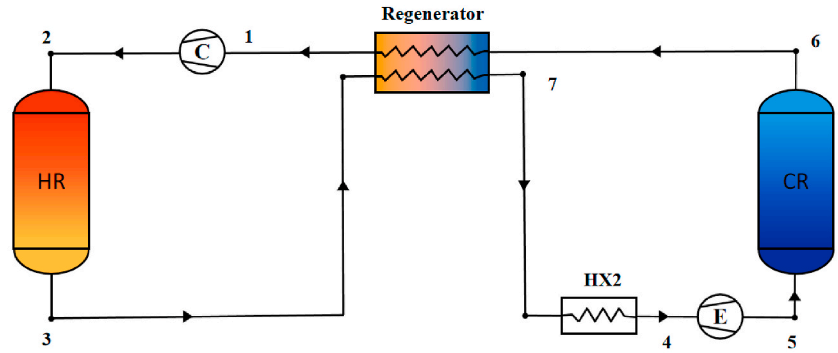
The packed bed thermal energy storage technology is regarded as the most appropriate for Brayton cycle PTES systems owing to its broad operating temperature range, substantial specific surface area and cost-effectiveness [36]. Throughout the charging and discharging processes of the packed bed, a thermocline develops, within which the majority of the gas-solid temperature differential and heat transfer take place. Outside the thermocline, the temperature difference between the gas and solid is minimal. Consequently, the propagation of the thermocline can be conceptualized as a "counterflow heat exchanger" that moves back and forth, as shown in Fig. 4, with the gas and solid serving as two distinct heat transfer "fluids".

The average inlet and outlet temperatures of the hot reservoir (HR) during the charging process are presented as follows [18],

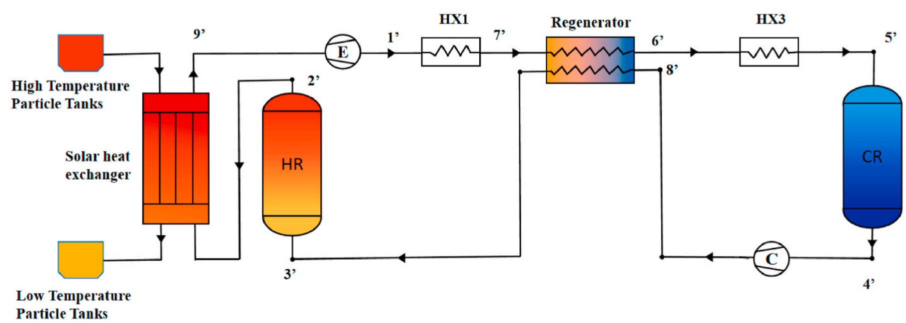
$$\Delta T_{\mathrm{HR,HT}} = \frac{\int_{t_0}^{t_{\mathrm{end}}} \left(T_{\mathrm{HR,g,in}} - T_{\mathrm{HR,s,out}}\right) dt}{t_{\mathrm{chr,end}} - t_{\mathrm{chr,0}}} \tag{3}$$

$$\Delta T_{\mathrm{HR,LT}} = \frac{\int_{t_0}^{t_{\mathrm{end}}} \left(T_{\mathrm{HR,g,out}} - T_{\mathrm{HR,s,in}}\right) dt}{t_{\mathrm{end}} - t_0} \tag{4}$$

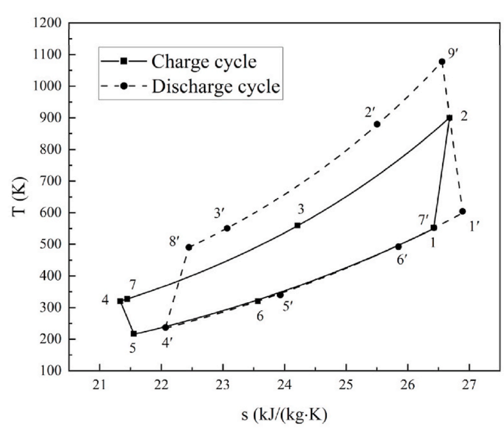
Wang et al. [18] developed a model for this hypothetical "counterflow heat exchanger" and determined that the heat capacity ratios of the solid (C_s) and gas (C_g) sides are approximately equal to 1. This balance guarantees that, during both the charging and discharging processes, the system operates at an equilibrium heat capacity ratio($(\dot{m}c_p)_g = (\dot{m}c_p)_{\mathrm{HR}} = (\dot{m}c_p)_{CR}$), thereby minimizing irreversible heat transfer losses. Under these conditions, the temperature differential between the hot and cold streams of the counterflow heat exchanger is uniform along its entire length, implying that the temperature differentials at the hot side ($\Delta T_{\mathrm{HR,HT}}$) and cold side ($\Delta T_{\mathrm{HR,LT}}$) of the packed bed are equal and in alignment with the ΔT_{HR} of the charging process. A similar situation occurs in the cold reservoir (CR) and in each tank during the discharge



(a) Charging process.



(b) Discharging process.



(c) T-s diagram.

Fig. 3. CSP-PTES system.

Table 1
System design parameters [18].

Parameter	T_{max} (K)	T_{env}(K)	ΔT_{CR}(K)	ΔT_{HR}(K)	p_1,$p_{1'}$(MPa)	$T_{s,in}$(K)
Value	800–900	300	10	10	0.105	1000–1200
Parameter	κ	φ	η_c, η_c'	η_e, η_e'	ρ_s(kg/m^3)	$c_{p,HR}$, $c_{p,CR}$(kJ/kg/K)
Value	0.4	0.4	0.9	0.9	5175	0.23 + 0.00201 T

process. Furthermore, pressure losses within the reservoirs will be represented by Δp. Specific parameters are shown in Table 2.

2.4. Solar heat exchanger

CSP plants utilize solar energy to heat particles to high temperatures and store them in high-temperature particle tanks for subsequent power generation. The solar-heated particles employed in this system are sourced from the CSP plant's high-temperature storage, since CSP plants can implement flow control strategies to maintain a constant temperature of heated particles stored in high-temperature tanks, assuming the tanks are adiabatic, the temperature of the particles entering the CSP-PTES system remains constant.

The solar heat exchanger is modeled with the heat exchanger effec-

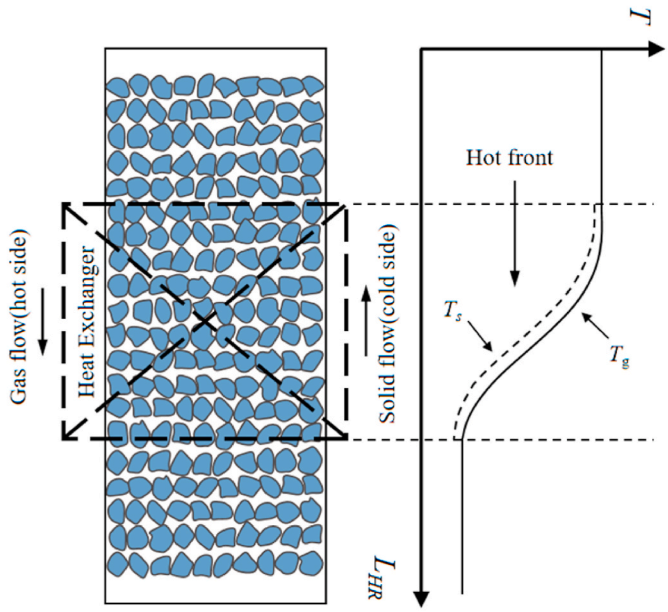


Fig. 4. Diagram of moving heat exchanger hypothesis for packed bed TES.

tiveness ε and the pressure loss coefficient ζ. The effectiveness, ε, is expressed as [17,35],

$$\varepsilon=\frac{\dot{Q}}{\dot{Q}_{\max}}=\frac{(\dot{m}c_p)_{\mathrm{g}}(T_{\mathrm{g,out}}-T_{\mathrm{g,in}})}{(\dot{m}c_p)_{\min}(T_{\mathrm{s,in}}-T_{\mathrm{g,in}})} \tag{5}$$

where $\dot{Q}$ is the actual heat transfer and $\dot{Q}_{max}$ is the maximum possible heat transfer in an infinitely large heat exchanger. T_{s} represents the temperature of the solar-heated particles.

Analogous to the thermal reservoirs, the particle heat exchanger functions under the assumption of an equilibrium heat capacity ratio, implying $(\dot{m}c_p)_{\mathrm{g}}=(\dot{m}c_p)_{\mathrm{s}}$. This implies that the temperature differential between the particles and the working fluid remains uniform throughout the heat exchanger, $(mc_p)_{\mathrm{s}}(T_{\mathrm{s,in}}-T_{\mathrm{s,out}})=(mc_p)_{\mathrm{g}}(T_{\mathrm{g,out}}-T_{\mathrm{g,in}})$. Given the minimal heat loss attributable to fluidizing air, it can be assumed that the heat efficiency of the heater is 100 % [37]. Consequently, the enthalpy change of the high-temperature particles is equivalent to the enthalpy change of the working gas.

The input exergy of the system can be expressed as [17],

$$E_{\mathrm{X}}=\dot{m}\cdot\Delta e_{\mathrm{X}}=\dot{m}\cdot(\Delta h-T_0\cdot\Delta s)=\dot{m}\cdot\bar{c}_p\left(T_{\mathrm{s,in}}-T_{\mathrm{s,out}}-T_{\mathrm{env}}\,ln\frac{T_{\mathrm{s,in}}}{T_{\mathrm{s,out}}}\right) \tag{6}$$

The pressure loss coefficient, ζ, of the particle heat exchanger is defined as the ratio of the pressure loss of the working fluid between the inlet and outlet of the heat exchanger to the inlet pressure, as follows [17,35],

$$\zeta=\frac{\Delta p}{p_{\mathrm{in}}} \tag{7}$$

The outlet pressure of the working fluid exiting the particle heat exchanger is given by,

$$p_{out}=(1-\zeta)p_{in} \tag{8}$$

2.5. Thermodynamic performance metrics

The performance of PTES systems is typically assessed using parameters such as round-trip efficiency χ, energy density ρ_{e}, and power density ρ_{p}. The mathematical models for these parameters are analyzed in detail as follows.

The round-trip efficiency, χ, is defined as the ratio of the net power output during the discharging process $W_{\mathrm{dis,net}}$ to the net power input during the charging process $W_{\mathrm{ch,net}}$, is obtained from Ref. [18],

$$\chi=\frac{W_{\mathrm{dis,net}}}{W_{\mathrm{ch,net}}} \tag{9}$$

For systems assisted by solar heat, the net input work also includes the exergy input from solar energy, E_{X}. Thus, the round-trip efficiency for the CSP-PTES system, $\chi_{\mathrm{CSP\text{-}PTES}}$, is defined as [29],

$$\chi_{\mathrm{CSP-PTES}}=\frac{W_{\mathrm{dis,net}}}{W_{\mathrm{ch,net}}+E_X}=\frac{(T_{9'}-T_{1'})-(T_{8'}-T_{4'})}{(T_2-T_1)-(T_4-T_5)+T_{\mathrm{s,in}}-T_{\mathrm{s,out}}-T_{\mathrm{env}}\,ln\frac{T_{\mathrm{s,in}}}{T_{\mathrm{s,out}}}} \tag{10}$$

During the charging process, the circulating fluid follows the sequence 1 → 2→3 → 7→4 → 5→6 → 1 in its working path, as shown in Fig. 3(c). The pressure at point 1 (p_1) remains constant at 0.105 MPa. The pressure losses in the HR and the CR are denoted as Δp_{HR} and Δp_{CR}, respectively. Assuming a pressure loss coefficient for the regenerator f_{p} of 1 %, an efficiency ε of 97 %, and a pressure loss coefficient for other external heat exchangers of 0.1 %, with negligible pressure loss in the pipes. The relationship between the compression ratio and the expansion ratio can be expressed as,

$$\beta_{\mathrm{e}}=\frac{\left(\beta_{\mathrm{c}}-\frac{\Delta p_{\mathrm{HR}}}{p_1}\right)\left(1-f_{\mathrm{p,re}}\right)\left(1-f_{\mathrm{p,ex}}\right)}{\frac{1}{\left(1-f_{\mathrm{p,re}}\right)}+\frac{\Delta p_{\mathrm{CR}}}{p_1}} \tag{11}$$

where $f_{\mathrm{p,re}}$ is the pressure loss coefficient of the regenerator, and $f_{\mathrm{p,ex}}$ is the pressure loss coefficient of external heat exchangers.

During the charging cycle, the gas temperature at the compressor outlet T_2 is the maximum temperature of the charging cycle, $T_{\max}$, which is as follows,

$$T_{\max}=T_1\cdot\beta_{\mathrm{c}}^{\kappa/\eta_c} \tag{12}$$

The corresponding gas temperature at the expander outlet T_5 represents the minimum temperature of the entire system operation $T_{\min}$, which is as follows,

$$T_{\min}=T_4\cdot\beta_{\mathrm{e}}^{-\kappa\eta_e} \tag{13}$$

The introduction of the regenerator significantly mitigates the temperature differentials within the reservoir, thereby enhancing the system's round-trip efficiency. The inlet temperature on the low-pressure side of the regenerator, T_6, is set equal to the expander inlet temperature, T_4. According to the thermodynamic model of the regenerator, the remaining inlet and outlet temperatures can be ascertained. The circulating fluid exchanges heat with the regenerator after absorbing and releasing heat within the two reservoirs. Consequently, the initial temperatures of the two reservoirs are influenced by the inlet temperatures of the regenerator. The initial temperature of the cold reservoir $T_{\mathrm{CR,ini}}$ is given by,

$$T_{\mathrm{CR,ini}}=T_6+\Delta T_{\mathrm{CR}} \tag{14}$$

The initial temperature of the hot reservoir $T_{\mathrm{HR,ini}}$ is,

Table 2
System loss parameters [18].

Parameter	ΔT_{HR}(K)	ΔT_{CR}(K)	ζ	Δp_{HR}(Mpa)	Δp_{CR}(Mpa)	$\Delta p'_{\mathrm{HR}}$(Mpa)	$\Delta p'_{\mathrm{CR}}$(Mpa)
Value	10	10	0.1 %	0.002	0.002	0.002	0.002

$$T_{\mathrm{HR,ini}} = T_3 - \Delta T_{\mathrm{HR}} \tag{15}$$

During the discharging process, the system operates as a heat engine, with the circulating fluid flowing in the reverse direction of the charging cycle. The operational path follows the sequence 9′→1′→7′→6′→5′→4′→8′→3′→2′→9′, as shown in Fig. 3(c). At point 1′, the pressure $p_{1'}$ remains constant, identical to the pressure at point 1 during charging, which is set at 0.105 MPa. The pressure losses in HR and CR are denoted by $\Delta p_{\mathrm{HR}'}$ and $\Delta p_{\mathrm{CR}'}$, respectively. Neglecting pipeline losses, the relationship between compression and expansion ratios during the discharging process is given as follows,

$$\beta_{\mathrm{c}}' = \frac{\frac{1}{1-\zeta}\beta_{\mathrm{e}}' + \frac{\Delta p_{\mathrm{HR}}'}{p_{1'}}}{\left(1 - f_{\mathrm{p,re}}\right)^2 \left(1 - f_{\mathrm{p,ex}}\right)^2 - \left(1 - f_{\mathrm{p,re}}\right)\frac{\Delta p_{\mathrm{CR}}'}{p_{1'}}} \tag{16}$$

During a single charge and discharge cycle, the circulating fluid engages in heat exchange with both HR and CR twice. For example, within the hot reservoir, the fluid releases heat during the charging phase and absorbs heat during the discharging phase. The temperature difference at the tank ports fluctuates twice, resulting in the discharging outlet gas temperature $T_{2'}$, as shown,

$$\rho_{\mathrm{e}} = \frac{\Delta e_{\mathrm{X}}}{V_{\mathrm{HR}} + V_{\mathrm{CR}}} = (1-\varphi)\frac{(mc_p)_{\mathrm{g}}\left[(T_2 - T_3) - T_{\mathrm{env}}\ln\left(\frac{T_2}{T_3}\right) + (T_5 - T_6) - T_{\mathrm{env}}\ln\left(\frac{T_5}{T_6}\right)\right]}{\left(\frac{m}{\rho}\right)_{\mathrm{HR}} + \left(\frac{m}{\rho}\right)_{\mathrm{CR}}} \tag{23}$$

$$T_{2'} = T_{\max} - \Delta T_{\mathrm{HR}} - \Delta T_{\mathrm{HR}}' \tag{17}$$

Similarly, the compressor inlet gas temperature $T_{4'}$ can be determined as,

$$T_{4'} = T_{\min} + \Delta T_{\mathrm{CR}} + \Delta T_{\mathrm{CR}}' \tag{18}$$

Based on the thermodynamic model of the solar heat exchanger, the gas temperature heated by the solar particles $T_{9'}$ is derived by,

$$T_{9'} = \varepsilon \cdot \left(T_{\mathrm{s,in}} - T_{2'}\right) + T_{2'} \tag{19}$$

Consequently, in accordance with the definition of round-trip efficiency, the expression for the round-trip efficiency is formulated as follows,

$$\chi = \frac{T_{9'}\cdot\left(1 - \beta_{\mathrm{e}}'^{-\kappa\eta_{\mathrm{e}}'}\right) - \left(T_{\min} + \Delta T_{\mathrm{CR}} + \Delta T_{\mathrm{CR}}'\right)\cdot\left(\beta_{\mathrm{c}}'^{\kappa/\eta_{\mathrm{c}}'} - 1\right)}{T_{\max}\cdot\left(1 - \beta_{\mathrm{c}}^{-\kappa/\eta_{\mathrm{c}}}\right) - T_4\cdot\left(1 - \beta_{\mathrm{e}}^{-\kappa\eta_{\mathrm{e}}}\right) + T_{\mathrm{s,in}} - T_{\mathrm{s,out}} - T_{\mathrm{env}}\, ln\frac{T_{\mathrm{s,in}}}{T_{\mathrm{s,out}}}} \tag{20}$$

Under ideal conditions, the internal pressure losses within the cycle are relatively minor and can be considered negligible. By setting $\frac{\partial\chi}{\partial\beta_{\mathrm{c}}'} = 0$, the optimal discharging compression ratio $\beta_{\mathrm{c,opt}'}$ can be derived as follows:

$$\beta_{\mathrm{c,opt}}' = \left(\frac{\eta_{\mathrm{c}}'\eta_{\mathrm{e}}'T_{9'}}{T_{\min} + \Delta T_{\mathrm{CR}} + \Delta T_{\mathrm{CR}}'}\right)^{\frac{1}{\kappa/\eta_{\mathrm{c}}' + \kappa\eta_{\mathrm{e}}'}} \tag{21}$$

The corresponding expansion ratio $\beta_{\mathrm{e}'}$ for discharging can be calculated from the above equation.

Energy density constitutes another critical metric for evaluating energy storage systems. It quantifies the amount of electrical energy that can be stored per unit volume or mass. In essence, it indicates the capacity to store electrical energy within a given volume or weight. A higher energy density implies that, for the same power output, the storage system can be more compact, thereby reducing footprint and space requirements. High energy density typically minimizes the size and weight of the equipment, thereby enhancing the usability and reliability of the energy storage system. With the same tank volume, a higher energy density results in more energy storage and a longer duration of power supply during operation.

The energy density ρ_{e} is defined as the ratio of the total electrical energy input during the charging period to the total volume of the energy storage device [18],

$$\rho_{\mathrm{e}} = \frac{W_{\mathrm{ch,net}}}{V_{\mathrm{HR}} + V_{\mathrm{CR}}} \tag{22}$$

The integration of a regenerator induces changes in the inlet and outlet temperature points of the reservoirs within the CSP-PTES system. This paper redefines energy density ρ_{e} as the ratio of the total exergy input to the total volume of the energy storage device to accurately characterize the system's energy density,

where φ is the porosity of the packed bed, and ρ is the density of the TES material.

Since the system operates with a balanced heat capacity ratio during charging, ρ_{e} can be further expressed as,

$$\rho_{\mathrm{e}} = (1-\varphi)\frac{(T_{\max} - T_3) + (T_5 - T_6) - T_{\mathrm{env}}\ln\left(\frac{T_{\max}}{T_3}\cdot\frac{T_5}{T_6}\right)}{\frac{1}{\rho c_{p,HR}} + \frac{1}{\rho c_{p,CR}}} \tag{24}$$

The power density represents another critical metric for evaluating the performance of energy storage systems. A higher power density indicates that the PTES system can perform more output work for the same working fluid volume, thereby meeting load demands more rapidly. Furthermore, for a given output power, a higher power density implies the utilization of more compact turbomachinery, which helps reduce the overall system footprint and enhance economic performance.

The power density ρ_{p} is defined as the ratio of the output power during the discharging process to the maximum volumetric flow rate of the working fluid in the discharging cycle [13],

$$\rho_{\mathrm{p}} = \frac{\dot{W}_{\mathrm{dis,net}}}{\dot{V}_{\max}} = \rho_{1'}c_{p,g}[(T_{9'} - T_{1'}) - (T_{8'} - T_{4'})] \tag{25}$$

where $\rho_{1'}$ denotes the density of the working fluid at the exit of the expander at point 1'. After expansion, the working fluid's volumetric flow rate is highest and its density is lowest at point 1'.

Based on the definition of power density, the mathematical expression for power density ρ_{p} can be formulated as,

$$\rho_{\mathrm{p}} = \frac{p_{1'}M_{He}c_{p,g}\cdot\left[T_{9'}\cdot\left(1 - \beta_{\mathrm{e}}'^{-\kappa\eta_{\mathrm{e}}'}\right) - \left(T_{\min} + \Delta T_{\mathrm{CR}} + \Delta T_{\mathrm{CR}}'\right)\cdot\left(\beta_{\mathrm{c}}'^{\kappa/\eta_{\mathrm{c}}'} - 1\right)\right]}{R\cdot T_{9'}\beta_{\mathrm{e}}'^{-\kappa\eta_{\mathrm{e}}'}} \tag{26}$$

where M_{He} is the molar mass of helium and R represents the molar gas constant.

The system performance parameters also include the solar efficiency, which is the efficiency of the thermoelectric conversion of solar energy. It represents the ability of the system to convert solar heat into electricity. In this paper, the solar efficiency of the CSP-PTES system is defined as the difference between the net power output of the CSP-PTES system and the net power output of the PTES system without solar heating over the solar heat input. The definition is provided as follows [28]:

$$\eta_{\text{solar}} = \frac{W_{\text{CSP-PTES,dis}} - W_{\text{PTES,dis}}}{\Delta E_{\text{solar}}} \tag{27}$$

where $W_{\text{CSP-PTES,dis}}$ is the net power output of the PTES system with additional solar input, and $W_{\text{PTES,dis}}$ is the net power output of the PTES system without solar input.

The thermodynamic models presented in this study are validated by replicating the calculations of a system proposed in the literature [18]. A comparison between the calculated results and those reported in the literature is provided in Table 3. The results are essentially consistent and therefore the reliability of the model can be proven.

All results presented in this study were obtained using MATLAB R2022.

3. Parameters impact analysis

The round-trip efficiency, $\chi_{\text{CSP-PTES}}$, energy storage density, ρ_e, and power density, ρ_p, of the CSP-PTES system are predominantly influenced by parameters such as the charging compression ratio, β_c, discharging compression ratio, β'_c, turbomachinery polytropic efficiency, η, effectiveness, ε, and pressure loss coefficient, ζ, of the solar heat exchanger, the inlet temperature of high-temperature particles, $T_{s,in}$, the maximum temperature during the charging cycle, T_{max}, and the environmental temperature, T_{env}. Among these parameters,η, ε, and ζ are the key characteristics, whereas β_c, β'_c, $T_{s,in}$, T_{max} and T_{env} are the system cycle state parameters.

During the modeling process described in Section 2, the relationship between the optimal discharging compression ratio and the charging compression ratio was established, as illustrated in Fig. 5. As β_c increases, β'_c also increases, but at a slower rate than that of β_c.

Fig. 6(a) illustrates the impact of the charging compression ratio β_c on $\chi_{\text{CSP-PTES}}$ under varying maximum charging temperatures T_{max}. As β_c increases, $\chi_{\text{CSP-PTES}}$ initially decreases and then slightly increases. Moreover, $\chi_{\text{CSP-PTES}}$ improves as T_{max} rises.

As the maximum charging temperature T_{max} increases, it makes the temperature of the work gas entering the solar heat exchanger during the discharge phase increase, resulting in a decrease in the solar exergy input, which in turn leads to a decrease in the total input energy to the system. Based on that, $\chi_{\text{CSP-PTES}}$ increases with the increase in T_{max}.

When the maximum charging temperature T_{max} is held constant, a small β_c results in minimal energy input from the compressor, with the majority of energy supplied by the solar heat. As β_c increases, both the compression and expansion work during the charging and discharging processes rise, leading to increased compression and expansion losses per unit mass. This initially reduces $\chi_{\text{CSP-PTES}}$.

However, as shown in Fig. 5, as β_c continues to increase, the growth rate of β'_c in discharging process is smaller than that of β_c in charging process, which implies that the compression and expansion losses in the discharging process will be lower than that in charging process, and this indicates that the net energy of the system's discharging process grows

Table 3
Summary of model verification results.

		Literature [18]	Model
Working fluid		helium	helium
T_{max}	K	1100	1100
T_{env}	K	300	300
P_1	bar	1.05	1.05
β_c	–	5	5
β_c'	–	6	6
Round-trip efficiency	%	59.0	59.2
Isentropic efficiency	%		90.0
Pressure loss	bar		0.02
ΔT	°C		10

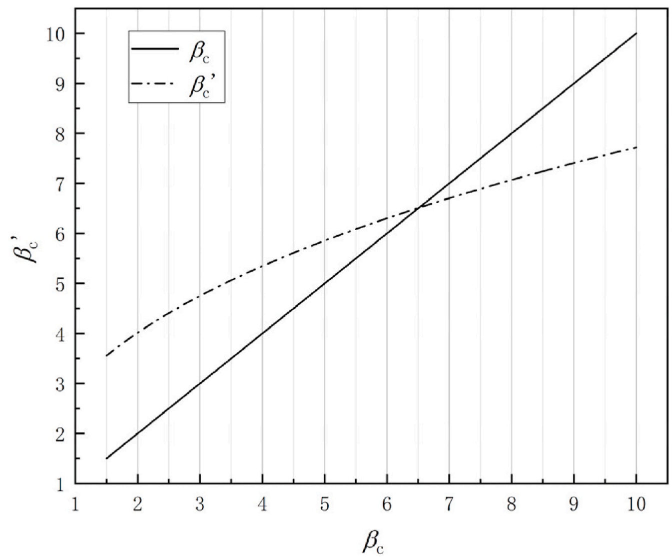


Fig. 5. Variation of optimal discharging compression ratio with charging compression ratio.

at a greater rate than that of the charging process, which leads to a slight increase in the $\chi_{\text{CSP-PTES}}$ at the later stages.

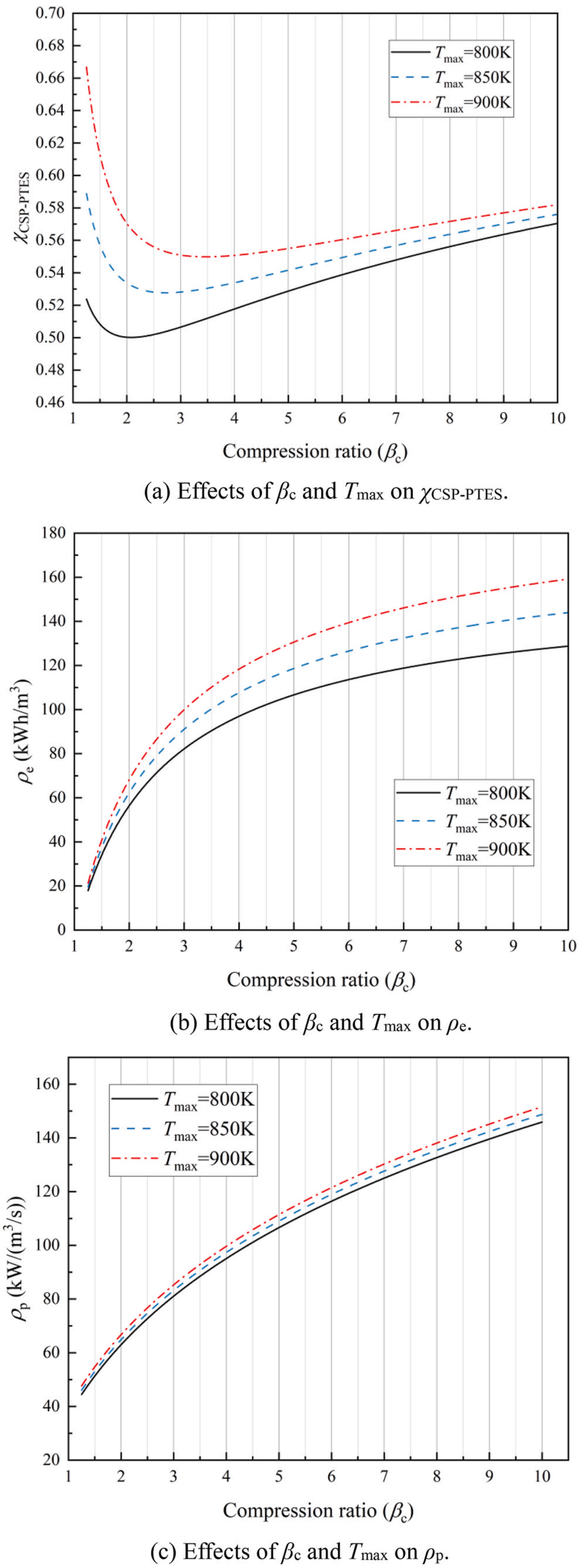
Fig. 6(b) shows the effect of β_c on ρ_e for different T_{max}. As β_c increases, ρ_e correspondingly increases. Furthermore, ρ_e rises with increasing T_{max}. Given the determined properties of the TES material and the porosity of the packed bed, the terms $(\frac{1}{\rho c_{p,HR}} + \frac{1}{\rho c_{p,CR}})$ and $(1-\varphi)$ related to these remain constant, indicating that ρ_e is influenced solely by other parameters. When T_{max} is constant, an increase in β_c lowers the compressor inlet temperature during the charging process. This means that the temperature difference between the entrance and exit of the hot reservoir will increase, so the exergy input into the reservoir will increase, the situation is similar for the cold reservoir. Combined, these factors result in an increase in ρ_e with rising β_c. When T_{max} increases, an increase in the heat storage capacity of the hot reservoir means that the exergy input into the reservoir will also increase, resulting in an increase in ρ_e.

Fig. 6(c) illustrates the effect of β_c on ρ_p for different T_{max}. When T_{max} is constant, ρ_p increases as β_c rises. As T_{max} increases, ρ_p increases slightly, though the overall change remains minimal. As β_c gradually increases, the net output power during the discharging process also increases. The lower temperature at the expander outlet (point 1′) results in a higher gas density, thereby leading to an increase in ρ_p. As T_{max} increases, the temperature of the work gas entering the expander after being heated by the solar heat exchanger increases slightly, resulting in a slight increase in the net power output, thereby slightly enhancing ρ_p.

The polytropic efficiency of turbomachine η is a critical parameter influencing performance metrics. Existing research indicates that PTES systems exhibit high sensitivity to η. Fig. 7 shows the influence of η on $\chi_{\text{CSP-PTES}}$, ρ_e, and ρ_p. As η increases, both $\chi_{\text{CSP-PTES}}$ and ρ_p rise. However, the change in energy density differs: at lower β_c, ρ_e decreases with increasing η, whereas at higher β_c, ρ_e increases with η, though the variations are minor in both cases.

As η improves, losses during the compression and expansion processes diminish, thereby raising the net power output during discharging, which boosts both $\chi_{\text{CSP-PTES}}$ and ρ_p. The results demonstrate that both $\chi_{\text{CSP-PTES}}$ and ρ_p are quite sensitive to changes in η. Specifically, an increase of 10 % in η results in an approximate 30 % increase in $\chi_{\text{CSP-PTE}}$, while ρ_p is approximately doubled.

The energy storage part of the system consists of two parts: hot storage and cold storage. The incorporation of the regenerator results in a correlation between the temperature at the turbomachinery inlet and



(a) Effects of β_c and T_{max} on $\chi_{CSP\text{-}PTES}$.

(b) Effects of β_c and T_{max} on ρ_e.

(c) Effects of β_c and T_{max} on ρ_p.

Fig. 6. Effects of β_c and T_{max} on system performance parameters.

outlet and the temperature of the individual storage tanks. Consequently, the trends observed in ρ_e vary with different values of η. As η increases, although an increase in the compressor inlet temperature decreases the hot storage capacity of the hot reservoirs, simultaneously the exergy loss of the hot storage process decreases. Conversely, a reduction in the expander outlet temperature enhances the cold storage capacity of the cold reservoirs, yet the exergy loss of the cold storage process rises. At lower β_c, the exergy loss increased of the cold storage process is larger than the reduction in the hot storage process, so ρ_e decreases accordingly. On the contrary, at higher β_c, the exergy loss reduction in the hot storage process is more significant, leading to an increase in ρ_e.

The solar heat exchanger is an essential component of the CSP-PTES system. Since only the working gas during the discharging process passes through the solar heat exchanger, its parameters, according to the evaluation metrics, only affect $\chi_{CSP-PTES}$ and ρ_p, but not ρ_e.

Fig. 8 illustrates the effects of the pressure loss coefficient ζ and efficiency ε of the solar heat exchanger on $\chi_{CSP-PTES}$ and ρ_p. As ε increases, ρ_p also rises, but $\chi_{CSP-PTES}$ declines. However, as ζ increases, both $\chi_{CSP-PTES}$ and ρ_p diminish. As ε increases, the heat transfer rate of the solar heat exchanger increases and the heat transfer losses decrease, resulting in higher output power and ρ_p. However, the increased solar heat input elevates the system's total energy input, resulting in a reduction in $\chi_{CSP-PTES}$. As ζ increases, the energy loss caused by the pressure loss in the heat exchanger increases, leading to a decrease in ρ_p and $\chi_{CSP-PTES}$.

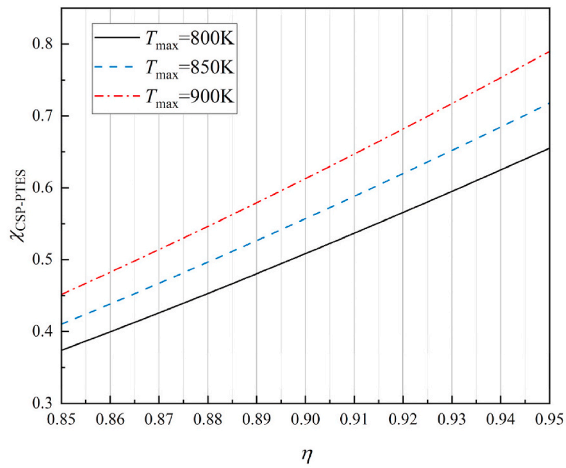
Fig. 9 shows the effect of the particle inlet temperature $T_{s,in}$ on $\chi_{CSP-PTES}$ and ρ_p. As $T_{s,in}$ increases, $\chi_{CSP-PTES}$ decreases, whereas ρ_p increases. As $T_{s,in}$ rises, the outlet temperature of the working gas heated by the high-temperature particles (expander inlet temperature) also increases, thereby enhancing the expansion power output and consequently increasing ρ_p. However, the total solar energy input to the system also increases, culminating in a decrease in $\chi_{CSP-PTES}$ due to the combined effects of these factors.

Environmental temperature is an essential factor to consider during system operation. Variations in environmental temperature have a significant impact on exergy loss during system operation and the design of state points, which in turn affects the overall performance of the system. Fig. 10 illustrates the effect of environmental temperature on $\chi_{CSP-PTES}$, ρ_e, and ρ_p. As T_{env} increases, $\chi_{CSP-PTES}$, ρ_e, and ρ_p all decrease. As T_{env} increases, exergy loss during the charging and discharging of the system also increases, resulting in a reduction in both ρ_e and $\chi_{CSP-PTES}$. Concurrently, the power output of the system declines, leading to a corresponding decrease in ρ_p.

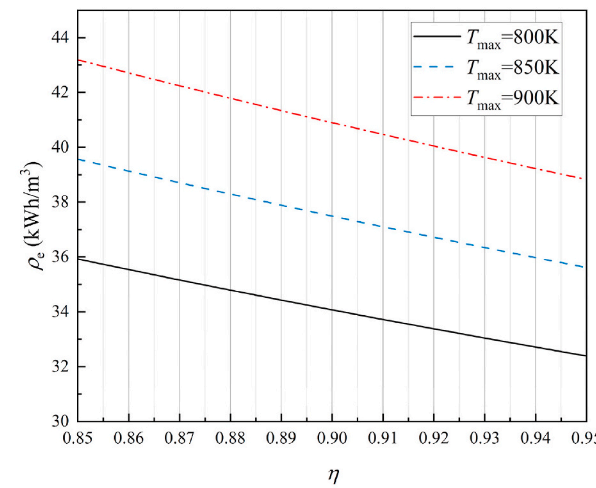
Fig. 11 illustrates the impact of solar heat input on the energy charging and discharging per unit mass flow of the CSP-PTES system. The input of solar thermal energy markedly boosts the electric power output per unit mass flow of the system while also substantially increasing the total energy input per unit mass flow of the CSP-PTES system.

As $T_{s,in}$ is fixed, the solar energy input per unit mass flow ΔE_{solar} remains constant. As shown in Fig. 11(a), at lower β_c, the PTES system without solar heating exhibits a greater power output growth rate than the CSP-PTES system. This is due to the fact that at lower β_c, the system energy is mainly supplied by solar energy input, with the energy input through the heat pump being relatively minor. Conversely, as the β_c increases, the energy input from the heat pump increases rapidly, resulting in a lesser percentage of the power output generated by the solar energy, which results in a decline in η_{solar}. In contrast, at larger β_c, the power output of the PTES system grows slowly due to increased compression and expansion losses. Concurrently, the percentage of power output generated by solar energy gradually rises, resulting in an increase in η_{solar}.

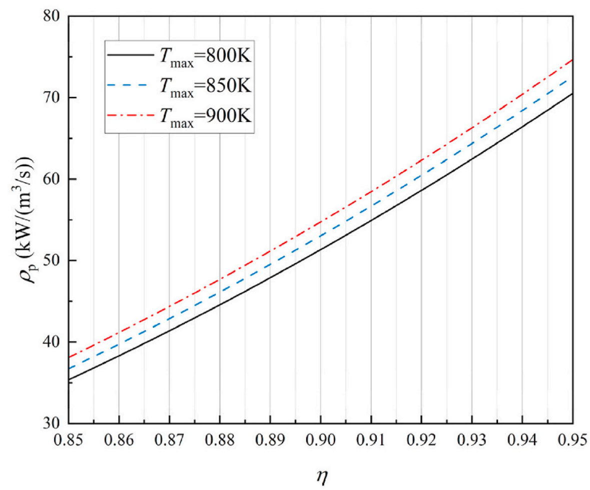
As $T_{s,in}$ grows, as shown in Fig. 11(a), an increase in $T_{s,in}$ does not significantly improve the net power output for smaller β_c. Instead, it markedly elevates the solar heat input, resulting in a lower η_{solar}.



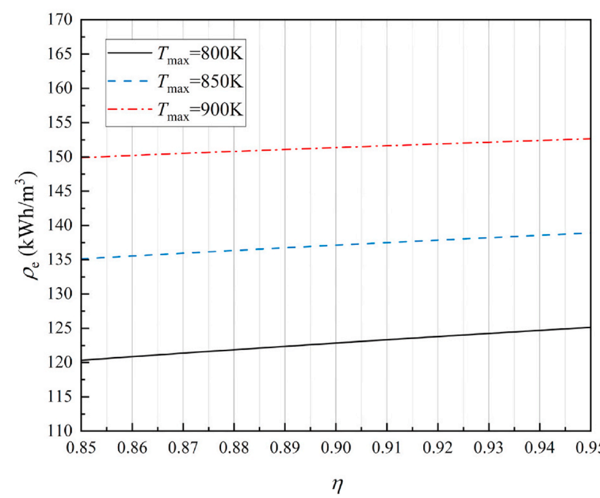
(a) Effects of η on $\chi_{CSP\text{-}PTES}$.



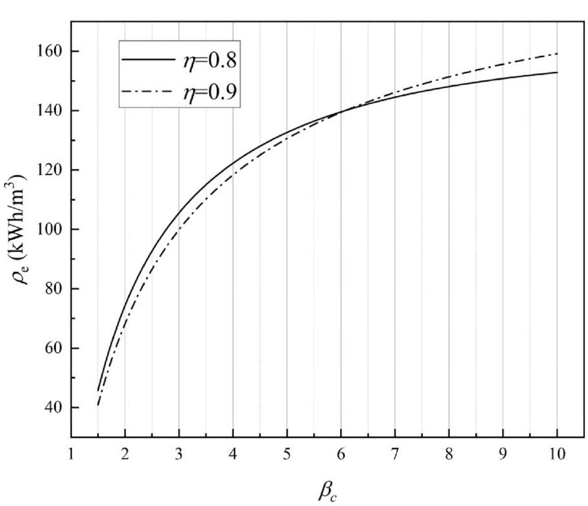
(d) Effects of η on ρ_e when $\beta_c = 1.5$.



(b) Effects of η on ρ_p.

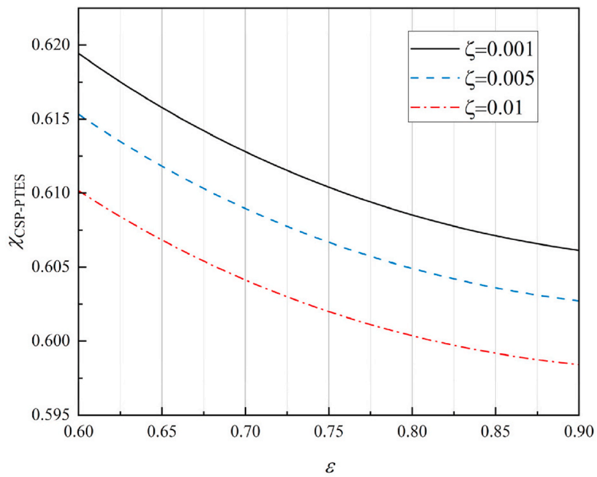


(e) Effects of η on ρ_e when $\beta_c = 8$.

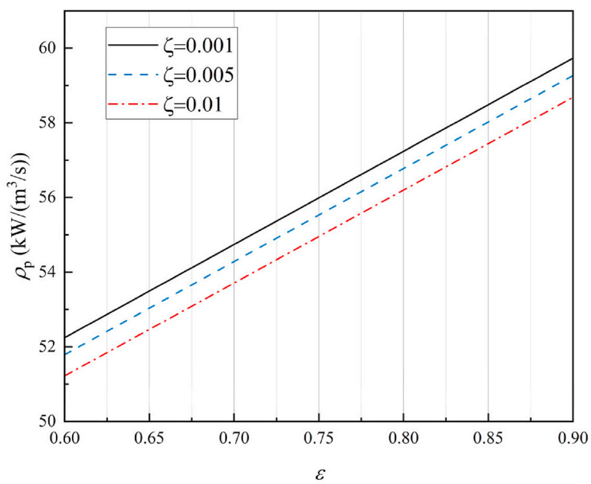


(c) Effects of η on ρ_e for different β_c.

Fig. 7. Effects of η on system performance parameters.

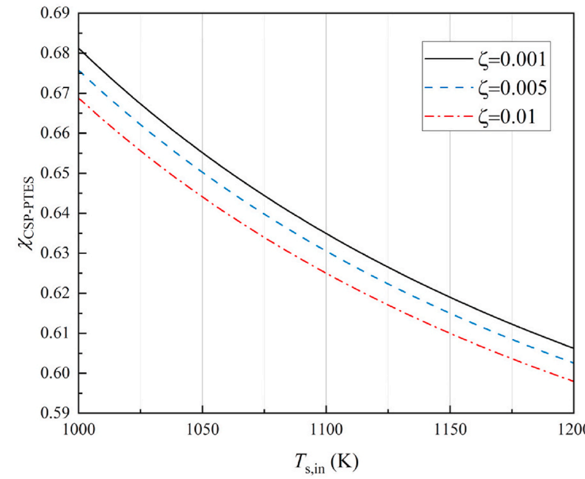


(a) Effects of ε and ζ on $\chi_{\text{CSP-PTES}}$.

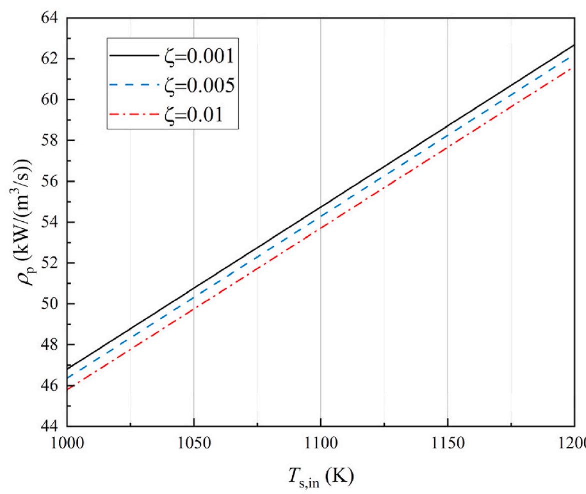


(b) Effects of ε and ζ on ρ_{p}.

Fig. 8. Effect of solar heat exchanger parameters on system performance parameters.



(a) Effects of $T_{\text{s,in}}$ on $\chi_{\text{CSP-PTES}}$.



(b) Effects of $T_{\text{s,in}}$ on ρ_{p}.

Fig. 9. Effect of high temperature particle inlet temperature on system performance parameters.

Conversely, for larger β_{c}, an increase in $T_{\text{s,in}}$ significantly enhance the net power output, thereby resulting in a greater η_{solar}.

When $\beta_{\text{c}} = 1.5$ (corresponding to a compressor temperature difference of about 130°C–150 °C, achievable with current compressor technology), η_{solar} can reach approximately 44 %. At higher β_{c}, η_{solar} can attain 44%–48 %, approaching the thermoelectric conversion efficiency of the s-CO2 Brayton cycle.

Fig. 12 shows the effect of the charging compression ratio β_{c} and the maximum particle temperature $T_{\text{s,in}}$ on the thermoelectric conversion efficiency η_{solar} of the solar heat exchanger. It can be observed that as β_{c} increases, η_{solar} initially decreases and then increases, similar to the trend of β_{c} on the round-trip efficiency $\chi_{\text{CSP-PTES}}$. For a fixed β_{c}, different $T_{\text{s,in}}$ values exhibit varying results: at lower β_{c}, higher $T_{\text{s,in}}$ corresponds to lower η_{solar}, although the overall difference is minor. On the contrary, at higher β_{c}, the higher $T_{\text{s,in}}$ corresponds to higher η_{solar}.

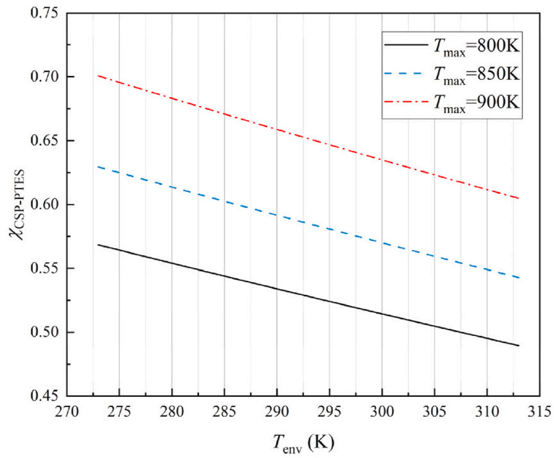
4. Multi-objective optimization and analysis of CSP-PTES systems

This section presents the numerical model and primary assumptions used to evaluate the comprehensive performance of the CSP-PTES system, along with the optimization and analysis of its performance metrics.

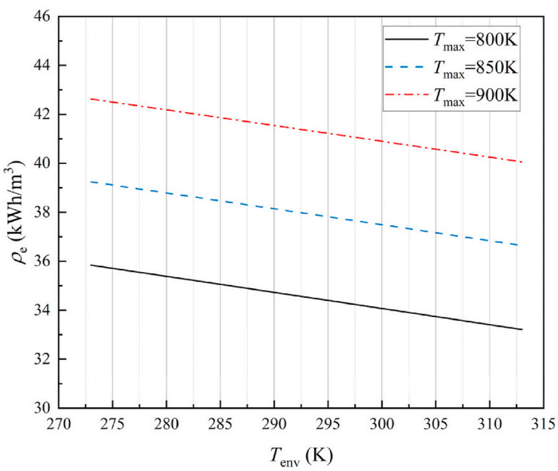
4.1. System numerical model

A numerical model for the CSP-PTES system is constructed, and the model code is developed in Matlab R2022. The thermal properties of helium are determined using REFPROP, while the thermophysical properties of basalt are obtained from Wang's literature [18]. The polytropic efficiency of the turbomachinery η, the efficiency ε and pressure loss coefficient ζ of the solar heat exchanger, and the pressure loss in the reservoirs are considered as fixed values under design conditions, consistent with the analysis in Section 2.

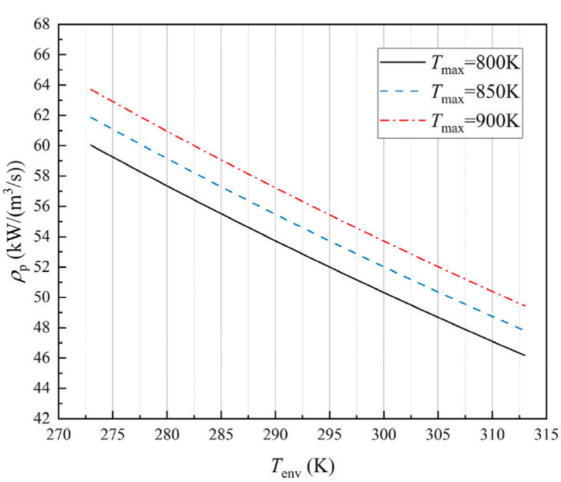
The turbomachinery is modeled utilizing a polytropic efficiency ($\eta_{\text{c}} = \eta_{\text{e}} = 0.9$), and the heat exchangers are modeled employing pressure loss coefficients and efficiencies. The assumptions included a pressure loss coefficient f_{p} of 1 % and an efficiency ε of 97 % for the regenerator, a pressure loss coefficient ζ of 0.1 % and an efficiency ε of 70 % for the solar heat exchanger, and a pressure loss coefficient f_{p} of 0.1 % and an efficiency ε of 90 % for other external heat exchangers.



(a) Effects of T_{env} on $\chi_{CSP\text{-}PTES}$.

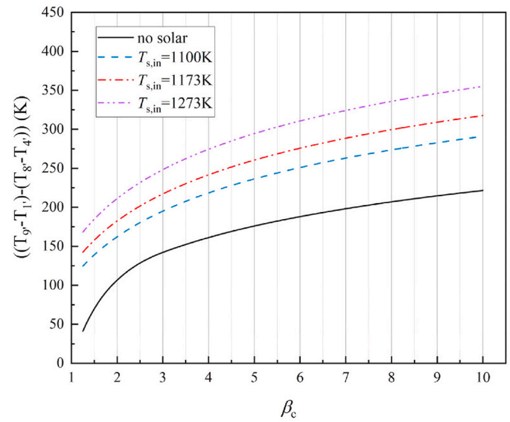


(b) Effects of T_{env} on ρ_e.

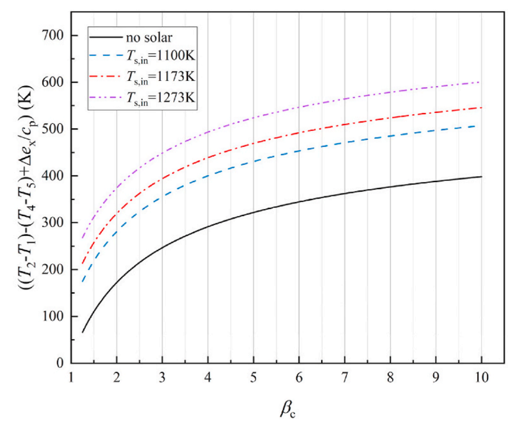


(c) Effects of T_{env} on ρ_p.

Fig. 10. Effect of environmental temperature on system performance parameters.



(a) Power output per unit mass flow rate.



(b) Sum of electric power input and solar exergy input unit mass flow rate.

Fig. 11. System power input and output per unit mass flow rate.

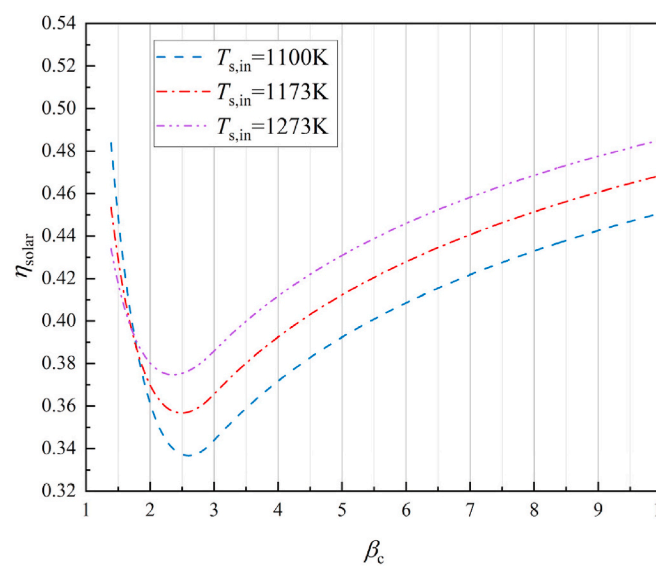


Fig. 12. Effect of system state parameters on thermoelectric conversion efficiency of solar heat exchanger.

Table 4
Energy equations of CSP-PTES system.

Heat pump cycle	Equations
Compressor	$W_{ch.c} = h_2 - h_1$
Expander	$W_{ch.e} = h_4 - h_5$
HR	$Q_{ch.HR} = h_2 - h_3$
CR	$Q_{ch.CR} = h_6 - h_5$
Regenerator	$h_3 - h_7 = h_1 - h_6$
Heat engine cycle	
Compressor	$W_{dis.c} = h_{8'} - h_{4'}$
Expander	$W_{dis.e} = h_{9'} - h_{1'}$
HR	$Q_{dis.HR} = h_{2'} - h_{3'}$
CR	$Q_{dis.CR} = h_{5'} - h_{4'}$
Solar heat exchanger	$Q_{dis.solar} = h_{9'} - h_{2'}$
Regenerator	$h_{7'} - h_{6'} = h_{3'} - h_{8'}$

Table 5
Component performance parameters and operating conditions of CSP-PTES system [18,31].

Fixed parameter	Unit	Values
Temperature of solar-heated particles	K	1100
Maximum temperature of heat pump cycle	K	900
Low pressure of heat pump cycles	MPa	0.105
Low pressure of heat engine cycle	MPa	0.105
Polytropic efficiency	–	0.9
Reservoir port temperature difference	K	10
Reservoir pressure loss	MPa	0.002
Decision variable parameter		
Maximum pressure of heat pump cycle	MPa	0.15–0.3
Maximum pressure of heat engine cycle	MPa	0.15–1

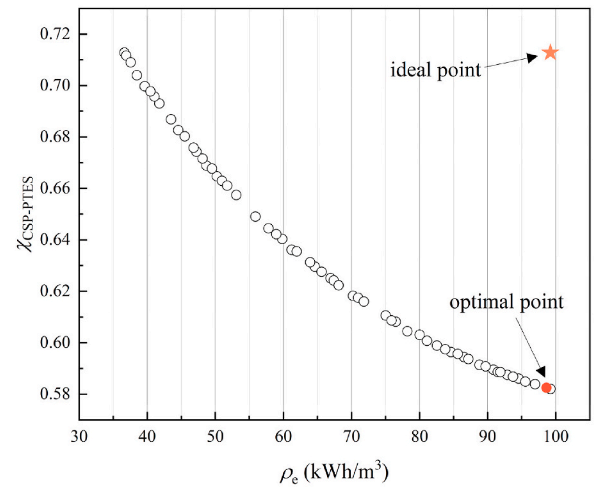
It is assumed that the system operates stably under design conditions, with negligible thermal leakage from the storage tanks and no pressure losses in the pipelines. The energy equations for the various components of the CSP-PTES system are shown in Table 4.

4.2. Multi-objective optimization and analysis

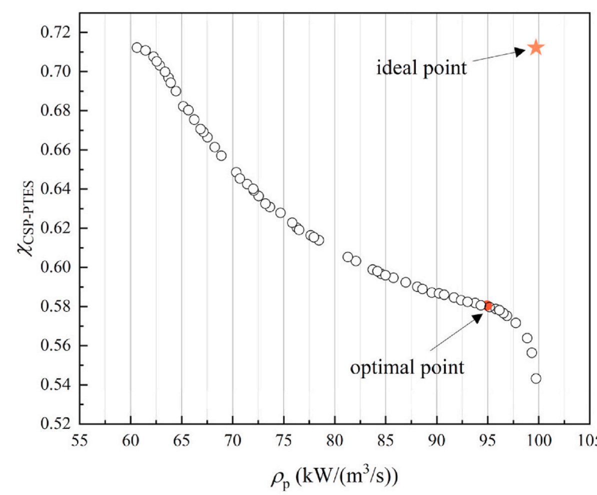
During the charging process, the minimum pressure in the CSP-PTES system, p_1(compressor inlet pressure), is set to 0.105 MPa, approximately equal to atmospheric pressure. The low pressure during the discharging process remains the same. The maximum charging temperature T_{max} is set at 900K, while the solar-heated particle inlet temperature $T_{s,in}$ is set at 1100K. Specific parameters are shown in Table 5.

This study conducts multi-objective optimization with round-trip efficiency $\chi_{CSP\text{-}PTES}$, energy storage density ρ_e, and power density ρ_p as the optimization objectives. For multi-objective genetic algorithm optimization problems, the Pareto front method of the Non-dominated Sorting Genetic Algorithm II (NSGA-II) is commonly utilized to present the optimal trade-off relationships among the optimization objectives [38]. The principle of NSGA-II involves defining the range of decision variables, selecting a set of points within this range using the algorithm, and iteratively comparing the performance of the corresponding objective functions through genetic iterations to ultimately determine the Pareto fronts between the optimization objectives. The solutions on the resulting Pareto front are non-dominated, meaning that if one objective function outperforms the corresponding value in other solutions, the other objective function must exhibit a trade-off.

The decision variables are selected as the maximum pressure during the charging cycle p_2 and the high pressure during the discharging cycle $p_{2'}$. The optimization objectives are to maximize the round-trip efficiency $\chi_{CSP\text{-}PTES}$, energy storage density ρ_e, and power density ρ_p. Fig. 13 presents the results of the 300-generation multi-objective optimization genetic algorithm, illustrating the Pareto fronts between $\chi_{CSP\text{-}PTES}$ and ρ_e, and between $\chi_{CSP\text{-}PTES}$ and ρ_p. The optimization aims to maximize the three objectives, with results indicating that $\chi_{CSP\text{-}PTES}$ decreases with an increase in ρ_e, and similarly decreases with an increase in ρ_p.



(a) The pareto front for $\chi_{CSP\text{-}PTES}$ and ρ_e.



(b) The pareto front for $\chi_{CSP\text{-}PTES}$ and ρ_p.

Fig. 13. Pareto fronts.

The Pareto front method of the Non-dominated Sorting Genetic Algorithm II (NSGA-II) produces a set of optimized solutions where no single solution is dominant over the others. Decision-makers typically select the final system design point from this set based on practical system requirements. This study employs the LINMAP (Linear Programming Techniques for Multidimensional Analysis of Preference) decision-making method [39], where the optimal value of each objective is designated as the "ideal point," as shown in Fig. 13. In multi-objective optimization, the "ideal point" typically does not

Table 6
State point parameters of the charge cycle for optimal decision point 1.

State point	T(K)	p(MPa)	h(kJ/kg)
1	566.43	0.105	2947.0
2	900.00	0.298	4679.9
3	574.67	0.296	2990.4
7	308.24	0.293	1606.8
4	300.00	0.292	1564.0
5	209.66	0.108	1094.3
6	300.00	0.106	1563.4
1	566.43	0.105	2947.0

Table 7
State point parameters of the discharge cycle for optimal decision point 1.

State point	T(K)	p(MPa)	h(kJ/kg)
1′	586.11	0.105	3049.2
9′	1034.00	0.508	5376.4
2′	880.00	0.509	4576.7
3′	563.62	0.511	2933.7
8′	472.53	0.516	2460.7
4′	229.66	0.102	1198.1
5′	320.00	0.104	1667.3
6′	475.34	0.104	2474.0
7′	566.43	0.105	2947.0
1′	586.11	0.105	3049.2

coincide with any point on the Pareto front. After normalizing the Pareto solutions and calculating the Euclidean distance between each solution and the ideal point, the solution exhibiting the smallest distance is chosen as the optimal decision point.

Within the range of decision variables, the variation of $\chi_{\text{CSP-PTES}}$ in Fig. 13(a) spans from 58.20% to 71.28 %, whereas ρ_e ranges from 36.64 kWh/m³ to 99.21 kWh/m³. The decision variables for the optimal decision point 1 of Fig. 13(a), determined using LINMAP, are $p_2 = 0.298$ MPa and $p_{2'} = 0.509$ MPa. The specific parameters for the charging and discharging cycle state points are detailed in Tables 6 and 7. The optimal state point is taken almost at the maximum energy density, achieving a round-trip efficiency of 58.25 %, an energy density of 98.62 kWh/m³, and a power density of 91.76 kW/(m³/s).

Within the same range of decision variables, $\chi_{\text{CSP-PTES}}$ in Fig. 13(b) varies from 54.33 % to 71.23 %, while ρ_p ranges from 60.62 kW/(m³/s) to 99.72 kW/(m³/s). The decision variables for the optimal decision point 2 in Fig. 13(b) are $p_2 = 0.299$ MPa and $p_{2'} = 0.560$ MPa. The corresponding optimal state point is likewise obtained at a higher power density, achieving a round-trip efficiency of 58.05 %, a power density of 94.93 kW/(m³/s), and an energy density of 98.98 kWh/m³. The specific parameters for the charging and discharging state points are detailed in Tables 8 and 9.

The two optimal state points in Fig. 13 are almost always obtained at low round-trip efficiencies, due to the fact that the addition of solar heat causes the round-trip efficiency of the CSP-PTES system to decrease and the output power to increase, leading to results presented on the Pareto fronts showing that the range of variation in round-trip efficiency would be smaller than the energy density and power density. According to the principle of the LINMAP decision-making method, the optimal point will be achieved at high energy density, high power density and low round-trip efficiency, but still achieving a round-trip efficiency of about 58 %.

A performance comparison between this system and other PTES systems integrated with CSP power plants is presented in Table 10. Despite some differences in parameter settings and working fluid selection, this study still demonstrates high round-trip efficiency compared with other researches. Additionally, the use of the packed bed enables the CSP-PTES system in this study to achieve significantly higher energy densities than those of liquid storage, with further design improvements potentially reaching energy densities exceeding 100 kWh/m³. Overall, the system presented in this study offers significant advantages in terms of overall performance.

Table 8
State point parameters of the charge cycle for optimal decision point 2.

State point	T(K)	p(MPa)	h(kJ/kg)
1	565.22	0.105	2940.8
2	900.00	0.299	4679.9
3	573.42	0.297	2984.0
7	308.20	0.294	1606.6
4	300.00	0.294	1564.1
5	209.29	0.108	1092.4
6	300.00	0.106	1563.4
1	565.22	0.105	2940.8

Table 9
State point parameters of the discharge cycle for optimal decision point 2.

State point	T(K)	p(MPa)	h(kJ/kg)
1′	566.02	0.105	2944.9
9′	1034.00	0.560	5376.5
2′	880.00	0.560	4576.8
3′	563.04	0.562	2930.9
8′	492.45	0.568	2564.4
4′	229.29	0.102	1196.2
5′	320.00	0.104	1667.3
6′	494.63	0.104	2574.2
7′	565.22	0.105	2940.8
1′	566.02	0.105	2944.9

Table 10
Performance comparison of PTES systems integrated with CSP power plants.

		Literature [28]	Literature [29]	Present study
Working fluid	–	nitrogen	argon	helium
Energy storage method		liquid storage	packed beds	packed beds
β_c	–	1.3–4.5	2–7	1.43–2.86
Maximum heat source temperature	K	1000	1000	1100
Isentropic efficiency	%	90	90	90
Round-trip efficiency	%	64–70	42–60	54–71
Energy density	kWh/m³	12–20	–	36–99

5. Conclusions

A novel PTES system (CSP-PTES) utilizing solar thermal energy from a CSP plant is proposed. The CSP-PTES system is designed with a power input of 10 MW, with both charging and discharging durations set at 4 h. Helium is selected as the working fluid, and a solar heat exchanger is integrated to heat the working fluid using high-temperature particles obtained from the CSP plant's collectors or high-temperature particle storage tank. A comprehensive parameter analysis and performance optimization were conducted for the CSP-PTES system utilizing solid packed beds. The primary conclusions are as follows.

(1) As the charging compression ratio β_c increases, the round-trip efficiency $\chi_{\text{CSP-PTES}}$ initially decreases and then slightly increases, whereas both the energy density ρ_e and power density ρ_p exhibit an upward trend. With an increase in the maximum charging temperature T_{max}, both $\chi_{\text{CSP-PTES}}$ and ρ_e increase, while ρ_p shows a slight improvement, though not significantly. As the polytropic efficiency η increases, both $\chi_{\text{CSP-PTES}}$ and ρ_p increase significantly. For energy density, at a low β_c, ρ_e decreases with increasing η, while at a high β_c, ρ_e increases with increasing η. However, the variations in ρ_e are minimal in both cases.

(2) As the efficiency ε of the solar heat exchanger increases, $\chi_{\text{CSP-PTES}}$ decreases, whereas ρ_p increases. Conversely, as the pressure loss coefficient ζ of the solar heat exchanger increases, both $\chi_{\text{CSP-PTES}}$ and ρ_p decrease, though the reduction is not significant. As the inlet temperature of solar-heated particles $T_{s,in}$ increases, $\chi_{\text{CSP-PTES}}$ decreases, but ρ_p increases. The solar heat input significantly enhances the charging and discharging energy of the system.

(3) As the environmental temperature T_{env} increases, $\chi_{\text{CSP-PTES}}$, ρ_e, and ρ_p all decrease. The variation in T_{env} has a more pronounced impact on $\chi_{\text{CSP-PTES}}$ and ρ_p, with ρ_e being less affected. As the charging compression ratio β_c increases, the thermoelectric conversion efficiency η_{solar} of the solar heat exchanger initially decreases and then increases, similar to the change in round-trip

efficiency. At both low and high β_c, the thermoelectric conversion efficiency of the solar heat exchanger can be up to 48 %, which is approaching the s-CO2 Brayton cycle. Additionally, at low β_c, a lower $T_{s,in}$ results in higher η_{solar}, whereas at high β_c, a higher $T_{s,in}$ results in higher η_{solar}.

(4) The performance optimization of the CSP-PTES system was investigated using a multi-objective optimization genetic algorithm to establish the trade-off relationships among $\chi_{CSP\text{-}PTES}$, ρ_e, and ρ_p. The results indicate that as $\chi_{CSP\text{-}PTES}$ increases, both ρ_e and ρ_p decrease accordingly. The LINMAP decision-making method identified two optimal points on the Pareto front. The addition of solar heat will lead to a decrease in the round-trip efficiency and an increase in the output power of the CSP-PTES system, resulting in a low round-trip efficiency of about 58 % at the optimal point. The performance metrics for optimal point 1 are $\chi_{CSP\text{-}PTES} = 58.25$ %, $\rho_e = 98.62$ kWh/m^3, and $\rho_p = 91.76$ kW/(m^3/s). For optimal point 2, the performance metrics are $\chi_{CSP\text{-}PTES} = 58.05$ %, $\rho_e = 98.98$ kWh/m^3, and $\rho_p = 94.93$ kW/(m^3/s).

The incorporation of solar thermal energy further elevates the working fluid temperature at the expander inlet, thereby enhancing the system's output power. The conversion efficiency of solar energy is notably enhanced, contributing to the increased capacity for renewable energy consumption and utilization. Although the round-trip efficiency experiences a slight reduction, it remains within acceptable limits, further validating the feasibility of incorporating solar heat into the PTES system. Furthermore, the system proposed in this study achieves significantly higher energy densities compared to other thermally-supported PTES systems, primarily due to the use of packed beds.

Nevertheless, there is still a lot of optimization space for the CSP-PTES system, and there is a certain amount of heat and pressure loss during the solar particle heat exchange process and heat storage in the solid packed beds, the optimization of the mentioned system components will further improve the system performance. Furthermore, considering the practical operating conditions, further investigation into the dynamic performance, start-stop capabilities, and operational flexibility of the CSP-PTES system is essential.

CRediT authorship contribution statement

Mengqi Du: Writing – original draft, Visualization, Investigation, Formal analysis, Data curation. **He Yang:** Validation, Investigation, Formal analysis. **Xiaoze Du:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization. **Hongwei Wu:** Writing – review & editing, Supervision, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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